COMP3751 VR/AR Coursework Report

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Abstract. This report summarises the coursework implementation from manual software rendering to IMU-based orientation tracking and simplified physics simulation. The strongest quantitative results are that complementary fusion reduces mean tilt error from 3.008 deg to 1.192 deg at the selected Problem 3 setting $\alpha = 0.20$, and that, in the reported magnetic-gain sweep, increasing the yaw-correction gain from $\beta = 0.02$ to $\beta = 0.20$ reduces mean post-correction heading error from 1.413 deg to 0.884 deg. The report also explains the scene design used for the submitted videos and the trade-offs between accuracy, smoothness, computational cost, magnetic-noise sensitivity, omitted magnetic calibration, and approximate sphere-based collision geometry.

1 Problem 1: Rendering

The rendering stage implements a manual rasterisation pipeline with model, view, projection, and viewport transforms, following the standard decomposition of the graphics pipeline into object, camera, projection, and viewport stages [1]. Objects are updated and redrawn every frame, so the scene is interactive rather than a single offline image. Multiple entities are supported by storing a mesh, colour, and transform for each object, and transforms are applied explicitly as $M = TRS$. This stage satisfies the requirement for real-time framebuffer display, multiple objects, and hand-written transformation matrices. Figure 1 shows a representative bunny render produced by the custom pipeline.

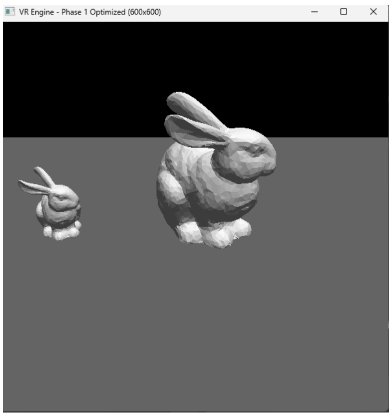


Figure 1: Representative Problem 1 bunny render produced by the custom software pipeline.

2 Problem 2: IMU Loading and Core Quaternion Operations

Problem 2 is limited to data loading and quaternion basics. The IMU CSV is read from file, gyroscope data are converted from degrees per second to radians per second, and the required quaternion operations are implemented manually: Euler $\leftrightarrow$ quaternion conversion, quaternion conjugation, and quaternion multiplication. These functions are the foundation for the later tracking stages. A short local validation script was used as a sanity check: the CSV loads as a 6959×10 array with monotonic timestamps, Euler $(0, 0, 0)$ maps to the identity quaternion $[1, 0, 0, 0]$, a sample quaternion multiplied by its conjugate returns identity, three Euler $\rightarrow$ quaternion $\rightarrow$ Euler round trips pass within a tolerance of 10^{-6}, and the integrated quaternion sequence remains normalised with maximum norm error 2.220×10^{-16}. Although additional helper operations were implemented later for rendering and filtering, they are not presented here as core Problem 2 deliverables.

3 Problem 3: Dead Reckoning and Complementary Fusion

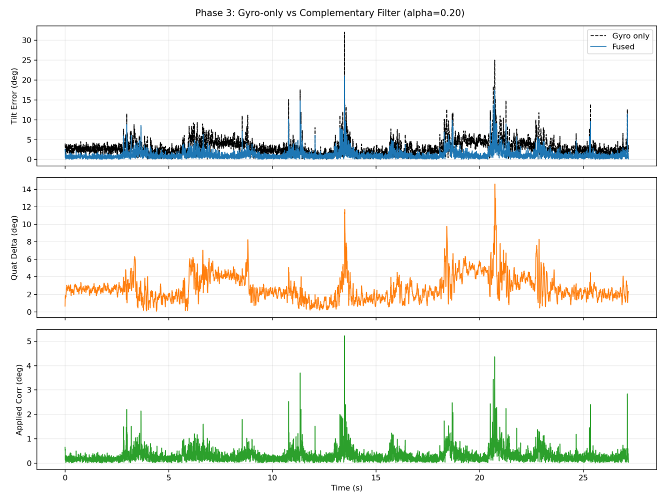


Figure 2: Problem 3 comparison using stable metrics rather than raw Euler angles at $\alpha = 0.20$: top, tilt error against gravity in degrees; middle, quaternion angular distance between fused and gyro-only trajectories in degrees; bottom, applied correction angle per frame in degrees.

Problem 3.1 uses gyro-only dead reckoning. Starting from the identity quaternion, each frame integrates angular velocity into a new orientation estimate. The weakness of this method is cumulative drift: small gyroscope error is integrated over time and is never corrected. Problem 3.2 adds accelerometer-based tilt correction through a complementary filter with gain α, following the standard IMU-fusion idea of combining high-frequency gyro integration with lower-frequency gravity alignment [2, 3]. In this report, *tilt error* is defined as the angle between the world-up vector and the accelerometer direction after rotating the normalised accelerometer reading into the world frame. Under this definition, the selected fused estimate at $\alpha = 0.20$ reduces mean tilt error from 3.008 deg to 1.192 deg, and reduces p95 tilt error from 6.241 deg to 3.138 deg. This corresponds to an improvement of about 60% in mean tilt error and about 50% at the p95 level relative to gyro-only. At the selected α, the final pose stays close to the gyro-only trajectory rather than diverging into

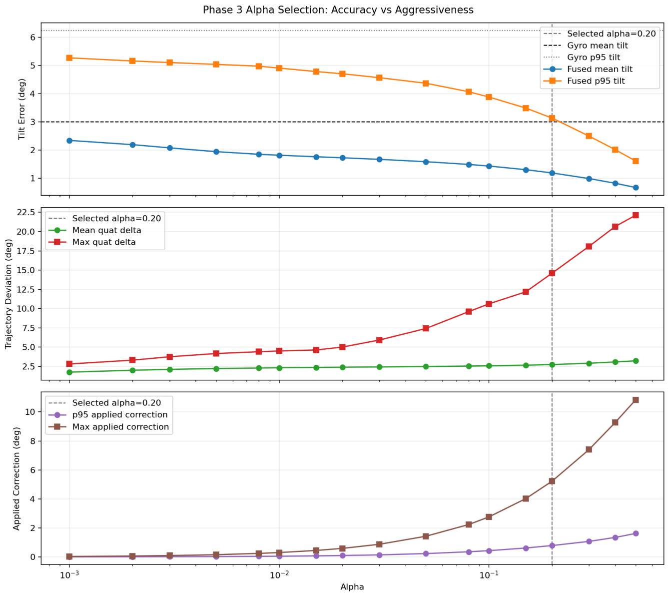


Figure 3: Problem 3 alpha selection trade-off with the final choice marked at $\alpha = 0.20$. Top: mean and p95 tilt error versus α, with gyro-only baselines. Middle: mean and maximum quaternion deviation from the gyro-only trajectory. Bottom: p95 and maximum applied correction step, which grows with α and matches the increasingly aggressive, visibly less smooth high-gain playback.

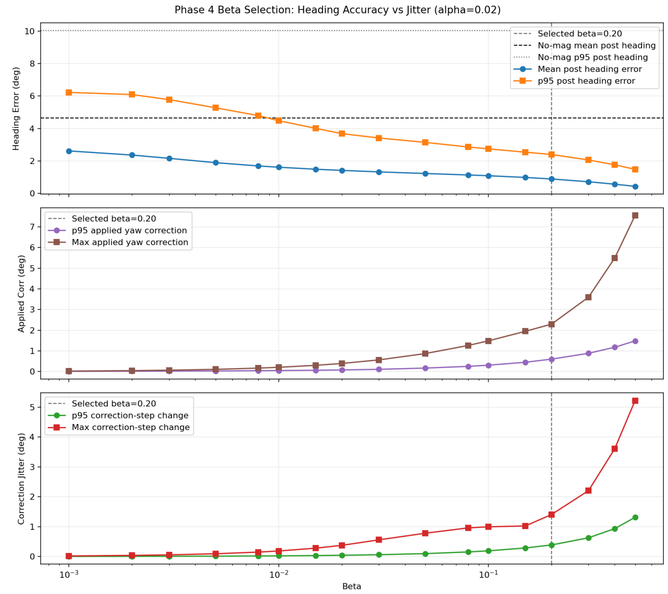


Figure 4: Problem 4 beta selection trade-off with the final choice marked at $\beta = 0.20$. Top: mean and p95 post-heading error versus β. Middle: p95 and maximum applied yaw correction. Bottom: p95 and maximum frame-to-frame correction change, used as a practical jitter proxy.

a qualitatively different motion. Figure 2 ties these claims to stable metrics: top, gravity alignment; middle, deviation from gyro-only; bottom, per-frame applied correction.

Problem 3.3 studies the choice of α as a parameter-selection problem rather than a one-metric optimisation. The extended sweep in Figure 3 shows that larger gains consistently reduce tilt error, but they also increase two costs: the quaternion deviation from the gyro-only trajectory and the magnitude of the applied per-frame correction. The bottom panel reports the p95 and maximum applied correction angle $\alpha\theta$, which is a useful proxy for how aggressively the filter perturbs the motion and is consistent with the visibly more wobbly playback observed at high α. Compared with the earlier conservative setting $\alpha = 0.02$, the adopted setting $\alpha = 0.20$ lowers mean tilt error by a further 31% (1.730 deg to 1.192 deg) and lowers p95 tilt error by a further 33% (4.709 deg to 3.138 deg), while increasing mean quaternion deviation from 2.391 deg to 2.738 deg and p95 applied correction from 0.096 deg to 0.784 deg. The strongest raw tilt result is obtained at $\alpha = 0.50$, but the gain in tilt accuracy is accompanied by much more aggressive correction, with maximum quaternion deviation rising to 22.115 deg and p95 applied correction to 1.617 deg. For this reason, $\alpha = 0.20$ was retained as the final Problem 3 setting: it captures a substantial additional improvement over $\alpha = 0.02$ without entering the visibly more wobbly high-gain regime. At this setting, the final yaw remains close to the gyro-only result (11.652 deg versus 12.233 deg), while the mean quaternion deviation is only 2.738 deg.

4 Problem 4: Magnetometer Yaw Correction

Problem 4 adds yaw correction from the magnetometer on top of the Problem 3 complementary filter. The current magnetic vector is rotated into the world frame, projected onto the horizontal plane, and compared against the initial horizontal magnetic reference. In this report, *heading error* is defined as the signed horizontal-plane angle between those two vectors. The resulting correction quaternion is applied about the world up-axis only, so roll and pitch are preserved while yaw is adjusted. Magnetometer vectors are normalised before comparison because the correction depends on direction rather than magnitude. Hard-iron and soft-iron calibration were not implemented; this remains a stated limitation rather than an error in the core yaw-correction logic.

The coursework text refers to trying different “alpha” values in this stage; in this implementation the accelerometer complementary gain remains denoted by α, while the magnetometer yaw gain is denoted by β to keep the two correction stages distinct. For Problem 4 the more conservative accelerometer setting $\alpha = 0.02$ is retained so that the beta sweep isolates the effect of magnetic yaw correction rather than re-tuning both parameters simultaneously. This full IMU stage is still conceptually a complementary fusion process, now with an extra heading-correction channel from the magnetometer [2, 3]. Figure 4 shows the resulting selection trade-off. Increasing β reduces post-correction heading error strongly relative to the no-magnetometer baseline, whose mean and p95 post-heading errors are 4.647 deg and 10.043 deg. At the selected runtime setting $(\alpha, \beta) = (0.02, 0.20)$ these global sweep metrics drop to 0.884 deg and 2.396 deg. This no-magnetometer baseline comparison is a global beta-sweep view. The later pre/post numbers quoted for the selected setting use a different local view: they compare heading error immediately before and after applying the magnetic correction within the same $\beta = 0.20$ run. However, larger β also increases both the applied yaw-correction size and the frame-to-frame change in that correction, which is a practical proxy for visible yaw jitter.

This is why $\beta = 0.20$ was retained for runtime playback rather than pushed to the numerically strongest setting. Relative to $\beta = 0.10$, it gives a clear extra accuracy

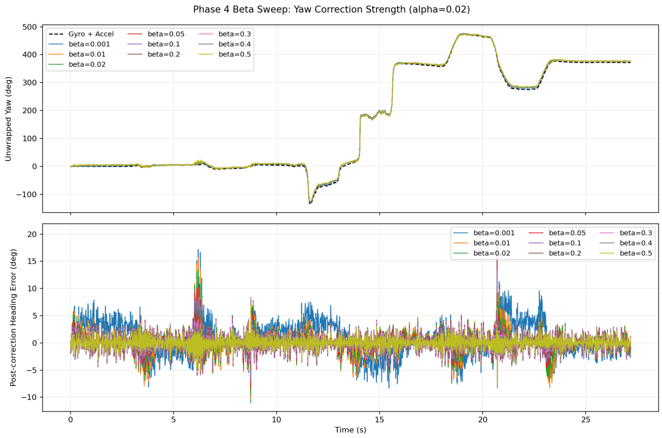


Figure 5: Problem 4 time-series beta sweep. Top: unwrapped yaw trajectories for gyro+accelerometer and full fusion under different beta values. Bottom: post-correction heading error over time. Stronger magnetic correction narrows the error band but also produces more visibly jittery temporal behaviour.

gain, reducing mean post-heading error from 1.084 deg to 0.884 deg. Pushing further to $\beta = 0.30$ lowers mean post-heading error again to 0.713 deg, but raises the p95 correction-step change from 0.387 deg to 0.626 deg. In that sense, $\beta = 0.20$ acts as the knee of the trade-off curve: it is stronger than $\beta = 0.10$, but avoids the sharper rise in correction jitter seen from $\beta = 0.30$ upward. The same pattern continues at higher gains. Moving from $\beta = 0.20$ to $\beta = 0.40$ lowers mean post-heading error from 0.884 deg to 0.562 deg, but nearly doubles the p95 applied yaw correction from 0.599 deg to 1.174 deg and raises the p95 correction-step change from 0.387 deg to 0.932 deg. At $\beta = 0.50$ the raw heading-error metric is strongest, but the p95 correction-step change reaches 1.311 deg, matching the visibly more twitchy motion in the higher-gain runtime video. The local baseline check at $\beta = 0.02$ is still positive, reducing mean absolute heading error from 1.442 deg to 1.413 deg. Under the local pre/post view of the chosen runtime setting, the mean absolute heading error drops from 1.106 deg before magnetic correction to 0.884 deg after correction, and the final-frame heading error drops from 0.987 deg to 0.790 deg.

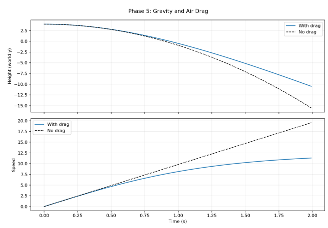
Figure 5 provides the complementary time-series view behind these summary metrics. The upper panel shows that all beta settings follow the same large-scale yaw trend, so the magnetometer is correcting drift rather than replacing the entire motion path. The lower panel shows the corresponding post-correction heading-error traces over time. Higher beta values narrow the error band, especially through the larger turns in the middle of the sequence, but also produce sharper, more frequent fluctuations, consistent with the correction-step statistics in Figure 4.

5 Problem 5: Gravity, Drag, and Simplified Collisions

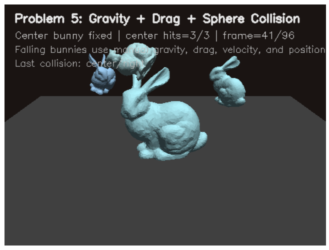
Problem 5 implements a lightweight manual physics model. Each dynamic body is updated by gravity and quadratic drag,

$$\mathbf{F}_{\text{drag}} = -\tfrac{1}{2}\rho C_d A \|\mathbf{v}\|^2 \hat{\mathbf{v}},$$

where the negative sign indicates that drag opposes the velocity direction. The current constants are $\rho = 1.3$, $C_d =$



(a) Drag versus no-drag motion.



(b) Physics scene after impacts.

Figure 6: Problem 5 evidence. Left: air drag reduces fall speed relative to the no-drag baseline. Right: the rendered physics scene shows the fixed center bunny and the three dynamic bunnies after collision.

0.5, and $A = 0.2$, following the standard quadratic-drag form used in the provided air-resistance reference [4]. Per frame the update is performed as $\mathbf{v}_{t+1} = \mathbf{v}_t + \mathbf{a}_t \Delta t$ and $\mathbf{p}_{t+1} = \mathbf{p}_t + \mathbf{v}_{t+1}\Delta t$. In the current run, drag reduces final fall speed from 19.538 to 11.317, and the left panel of Figure 6 shows the resulting separation between the drag and no-drag trajectories.

Collision detection is deliberately simplified to sphere overlap using the sum of radii. The center bunny uses radius 1.05, while the three moving bunnies use radii in the range 0.56–0.60; these values were chosen empirically so that the spheres roughly cover the scaled bunny bodies after scene scaling, while still delaying contact until the visible meshes are close enough for the collision to read plausibly on screen and the center bunny remains easy to hit. Collision response reflects the incoming normal velocity component; for floor response the velocity is decomposed into normal and tangential parts, $\mathbf{v} = \mathbf{v}_n + \mathbf{v}_t$, then updated as $\mathbf{v}'_n = -e\mathbf{v}_n$ and $\mathbf{v}'_t = d\mathbf{v}_t$ with restitution $e = 0.32$ and tangential damping $d = 0.97$. This model is intentionally approximate, but it satisfies the requirement that several objects collide and change direction. In the local collision check, the incoming normal component changes from -2.0 to $+2.0$, which provides a direct numerical change-direction test rather than only a visual impression, and in the rendered scene all three dynamic bunnies ('left', 'rear', and 'right') collide with the center bunny, rebound away from it, and remain visually separated afterwards, as illustrated by the right panel of Figure 6.

This simplification is justified by the bunny model itself. The mesh contains 2503 vertices and 4968 faces, so exact triangle-level collision would be substantially more expensive than the current sphere test and would have to be performed for several moving bunnies every frame. More accurate alternatives would include multiple bounding spheres, AABB/OBB broad-phase tests, or convex hulls. These would reduce the false positives of the current one-sphere approximation, but would also increase the number of per-frame overlap tests, the amount of bookkeeping, and the narrow-phase geometric cost relative to the current model.

6 Problem 6: Evaluation and Direct Answers

6.1 Static renders and video evidence. The static renders, generated plots, submitted videos, and extracted keyframes together provide the required evidence. The FAQ split is followed explicitly: when running the code, the intended sequence is Problem 3.2 then Problem 4.1; in the submitted video, the sequence is Problem 3.1 then

(a) Problem 3.1

(b) Problem 3.2

Figure 7: Representative submitted-video keyframes using the shared central-bunny-with-collisions scene.

Metric	Value
P3.1 final yaw	12.233 deg
P3.2 final yaw	11.652 deg at $\alpha = 0.20$
P3.2 mean / p95 tilt	1.192 deg / 3.138 deg; gyro 3.008 deg / 6.241 deg
P3.2 mean / p95 quaternion delta	2.738 deg / 5.396 deg
P3.2 p95 / max corr.	0.784 deg / 5.225 deg; at $\alpha = 0.50$: 1.617 deg / 10.826 deg
P4.1 final yaw	15.702 deg at $\alpha = 0.02$, $\beta = 0.20$
P4.1 selected-run mean heading pre / post	1.106 deg / 0.884 deg
P4.1 selected-run final heading pre / post	0.987 deg / 0.790 deg
P4.1 p95 heading / corr.-change	2.396 deg / 0.387 deg; at $\beta = 0.40$: 1.760 deg / 0.932 deg
P5 final speed magnitude	19.538 (no drag) vs 11.317 (drag)

Table 1: Key numerical evidence cited in the Problem 3–5 discussion.

Problem 3.2. In all scenes, the center bunny remains above the floor while other bunnies fall and collide with it. Figure 7 shows representative keyframes from the two submitted Problem 3 sequences.

6.2 Dead reckoning versus fusion. The strongest evidence is not the raw Euler-angle plot but the stable-metric comparison in Figure 2, together with the summary values in Table 1. Raw Euler-angle traces can become misleading near orientation singularities, so the comparison here relies on quaternion and gravity-alignment metrics instead. Dead reckoning drifts because angular velocity is integrated without external correction. The complementary filter improves stability by re-aligning roll and pitch with gravity. Under the selected $\alpha = 0.20$ setting, mean / p95 tilt error fall from 3.008 deg / 6.241 deg in gyro-only tracking to 1.192 deg / 3.138 deg after fusion, while mean / p95 quaternion deviation remain only 2.738 deg / 5.396 deg. The alpha-selection plot in Figure 3 adds the second half of the argument: although larger α values can further reduce tilt error, they also increase both trajectory deviation and the size of the applied correction step, which explains why the high-gain playback looks more wobbly. The metric-to-claim link is therefore direct: lower tilt error supports better gravity alignment, small quaternion distance supports the claim that fusion improves stability without replacing the original motion path, and smaller applied correction steps support smoother visual behaviour.

6.3 Performance and number of calculations. All implemented tracking methods are $O(1)$ per frame, but the amount of arithmetic differs. Dead reckoning performs a quaternion update only. Complementary fusion additionally requires accelerometer normalisation, vector rotation, dot and cross products, an angle computation, and a correction quaternion. Full magnetic fusion adds

Method	Total time	Time / frame
Dead reckoning	167.144 ms	24.018 µs
Gyro + accel	3697.181 ms	531.281 µs
Full fusion	7256.567 ms	1042.760 µs

Table 2: Measured runtime evidence for Problem 6.3 on the same local machine, reported as median warm-start timings.

horizontal projection, signed heading computation, and a second correction quaternion. Table 2 reports the corresponding runtime evidence from repeated warm-start measurements on the same local machine. Dead reckoning is the cheapest because it performs one orientation update, gyro+accelerometer adds one gravity-alignment correction stage, and full fusion adds a second directional-correction stage for heading. The measured ordering therefore follows directly from the algorithmic structure rather than being an isolated benchmark result.

6.4 Could position tracking be implemented? Yes, in principle. One would first estimate orientation, rotate acceleration into a consistent frame, subtract gravity, integrate once to obtain velocity, and integrate again to obtain position. In practice, expected accuracy would be poor because double integration magnifies even small bias and noise very quickly. This makes long-term absolute position tracking unreliable in the present IMU-only setup.

6.5 Are sphere collisions reasonable, and how could they be improved? Yes, as a first approximation. They are cheap, visually clear, and sufficient for the coursework scene. However, they can trigger contact earlier than the real mesh would. More accurate methods such as multiple spheres, AABB/OBB hierarchies, convex hulls, or triangle-level narrow phase would better match the bunny geometry, but would increase per-frame cost, especially for a mesh with 2503 vertices and 4968 faces. A practical next step would be to replace the single sphere with a small set of spheres or a simple broad-phase hierarchy, which would reduce geometric false positives while still remaining far cheaper than full triangle-level collision on the bunny mesh.

7 Conclusion

The project achieves the required progression from custom rendering to IMU-driven tracking and scene-based presentation. The strongest confirmed outcomes are that accelerometer fusion improves tilt stability over gyro-only dead reckoning, magnetometer correction improves heading consistency, and the simplified physics scene provides the required falling and colliding bunnies. The remaining trade-offs are explicit: stronger correction gains are less smooth, magnetic calibration was not implemented, and sphere collisions are approximate rather than exact.

References

[1] *EE267 Notes: Graphics Pipeline*, course-provided graphics pipeline notes.

[2] S. M. LaValle, Y. Yershova, M. Katsev, and M. Antonov, "Head Tracking for the Oculus Rift," in *Proceedings of the IEEE International Conference on Robotics and Automation*, 2014.

[3] S. O. H. Madgwick, *An efficient orientation filter for inertial and inertial/magnetic sensor arrays*, internal report, University of Bristol, 2010.

[4] *Falling Object with Air Resistance*, course-provided reference note used for the quadratic drag model discussion.